

Model Building (Model Evaluation)

- R^2 squared --- compare qilishda
 - Model data setdagi o'zgarishlarni qancha darajada tushunganimni o'lchaydi.
 - MSE (Mean Squared Error) ---
 - ko'proq ixtirida
 - y -pred bilan y -actual o'rtasidagi kvadratik farqlarning o'rtachasini hisob.
- RMSE (kvadrat ildizi o'rtacha kvadratik xatolik)
 - MSE ning kvadrat ildizi --- tashqarida
- MAE (o'rtacha absolute xatolik) --- tashqarida.

R^2 - score.

No	y -actual	y -pred
1	3	2.5
2	4.5	4
3	5	6
4	6	5.5
5	7	8

$$y_i = 5 \text{ Act}$$

$$\hat{y}_i = 5 \text{ Predict}$$

$$\bar{y}_i = 5 \text{ Mean of Act.}$$

Caption

$$\text{MSE} = \frac{\sum_{i=1}^n (y_{\text{act}i} - y_{\text{pred}i})^2}{n}$$

$$\text{MSE } ① (3 - 2.5)^2 = 0.25$$

$$② (4.5 - 4)^2 = 0.25$$

$$③ (5 - 6)^2 = 1$$

$$④ (6 - 5.5)^2 = 0.25$$

$$⑤ (7 - 8)^2 = 1$$

$$⑥ 0.25 + 0.25 + 1 + 0.25 + 1 = 2.75$$

$$⑦ \frac{2.75}{5} = 0.55$$

$$\text{MSE} = 0.55$$

$$\text{RMSE} = \sqrt{\text{MSE}}$$

$$\text{RMSE} = \sqrt{0.55} \approx 0.74$$

$$\text{MAE} = \frac{\sum_{i=1}^n |y_{\text{act}i} - y_{\text{pred}i}|}{n}$$

$$\text{MAE} = \frac{0.5 + 0.5 + 1 + 0.5 + 1}{5} = \frac{3.5}{5} = 0.7$$

$$R^2 = 1 - \frac{\text{MSE} \cdot n}{\text{Total error}}$$

$$R^2 = 1 - \frac{\sum (y_{\text{act}i} - y_{\text{pred}i})^2}{\sum (y_{\text{act}i} - \bar{y}_i)^2}$$

$$\bar{y} = \frac{3+4.5+5+6+7}{5} = \frac{25.5}{5} = 5.1$$

$$R^2 = 1 - \frac{2.75}{\sum (y_i - \bar{y})^2}$$

$$\sum (y_i - \bar{y})^2 = (3-5.1)^2 + (4.5-5.1)^2 + (5-5.1)^2 + (6-5.1)^2 + (7-5.1)^2$$

$$= 4.41 + 0.36 + 0.01 + 0.81 + 3.61 = 9.2$$

$$\sum (y_i - \bar{y})^2 = 9.2$$

$$= 1 - \frac{2.75}{9.2} = 1 - 0.3 = 0.7$$

near Models

n Models - feature va target qiymatlar o'rtasidagi chiziqli aloqa bo'lgan faraz qiledi.

Agar ko'p foydalaniladigan model equation

so'ngiy maqsad: Actual va predicted natijalar o'rtasidagi farqni kamaytirish

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

